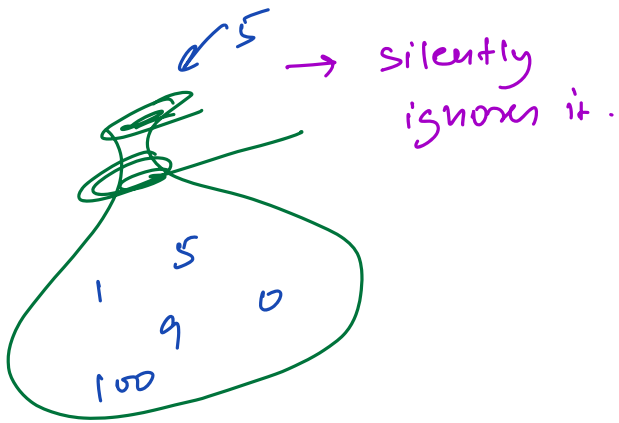


HashMap & HashSet

HashSet

collection of unique elements
↓ no ordering ↓ no duplicated allowed



Some examples of unique elements?

- email id
- username
- fingerprints

Example

array arr = [1, 3, -2, 7, 1, 1, -2]

HS = { 1, -2, 3, 7 }

Syntax

```
HashSet < Type > h = new HashSet < Type > ();
```

↓
any class
Integer, Double, ...

Basic Operations

Add : add element in hashset

Add : add element

Contains : check whether hashset contains certain element or not.

Size : total elements in hashtable

Remove : remove element from hashset

Print : we use for each loop to print elements
(not a function) of hash set

Analyst

Sequential Order

Duplicated Allowed

Wash Set

Sequence not maintained

Only unique elements

HashMap

data structure which contains key-value pairs.

Example

State (key)	Population (value)
Punjab	15
Haryana	18
UP	20
Delhi	5

(Haryana, 18)
(UP, 20)

UP → 20 is a pair
↓ ↘
key value

corresponding to which
some value is stored
and by this key we can
access that data.

examples of (key, value)

- user id - password
- word - meaning (dictionary)
- Item - price

Features of Hashmap

- Duplicate values are allowed

(UP, 20) (MP, 20)

- Duplicate keys are not allowed

(UP, 20) ~~(UP, 18)~~
 ~~(UP, 20)~~

- No order of data, key-value pairs are in random order.

Syntax

```
HashMap<KeyType, valueType> hm = new HashMap  
    <KeyType, valueType>();
```

Basic Operations

- add (K, V)
- contains (K)
- get (K) $\rightarrow$ (V)
- update (K, newVal)
- remove (K)
- print

[https://www.scaler.com/topics/java/online-java-compiler/?
snippet_slug=4c29602861314a6f4bdd](https://www.scaler.com/topics/java/online-java-compiler/?snippet_slug=4c29602861314a6f4bdd)